The impact of the US-China trade war on Vietnam's US Exports*

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Abstract

This paper examines the unintended consequences of the 2018–2019 US–China trade war for Vietnam, a key bystander economy, using quarterly HS 8-digit US import data from 2017Q1 to 2020Q4. While the average treatment effect on Vietnam's exports to the US is small and statistically insignificant, substantial heterogeneity arises from pre-war revealed comparative advantages (RCAs). Vietnam experienced markedly stronger export growth in sectors where China held high pre-war RCA, consistent with trade diversion and substitution effects. These gains were nonlinear and most pronounced in sectors where Vietnam initially had low comparative advantage, reflecting the exploitation of previously untapped export opportunities. Consequently, Vietnam captured meaningful trade-diversion benefits primarily outside its pre-existing areas of comparative strength. In addition, we provide consistent evidence that the relocation of MNCs is strongly associated with the observed export surge. By quantifying these spillovers, our study contributes to the literature on global value chain dynamics and trade protectionism.

Keywords: US-China Trade War; Vietnam; MNC
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1 Introduction

The tariffs imposed by the United States on Chinese imports in 2018 and 2019 have garnered significant attention from policymakers, media, and academia due to their profound impact on global trade. In 2018, the US government implemented tariffs on approximately $250 billion of Chinese imports, rolled out in three waves (July, August, and September), covering roughly 50% of US imports from China based on 2017 trade values. In September 2019, additional tariffs were imposed on $125 billion of Chinese goods, bringing the total to 72% of 2017 US imports from China. In retaliation, China imposed tariffs on 97% of US exports to China, escalating bilateral trade tensions. The US-China trade war was partially mitigated by the Phase One deal signed in January 2020, which reduced some tariffs but left most 2018–2019 tariffs intact. Protectionist trade policy has reemerged under the new Trump administration (Trump 2.0), with the introduction of various new tariffs. These developments underscore the urgent need for timely evidence on the broader consequences of the trade war and to inform future trade policy decisions.

This paper examines the trade war's impact on Vietnam, a key bystander, using its US exports as a case study to elucidate trade relocation mechanisms. Vietnam's proximity to China, shared institutional features, and low labor costs position it as a likely beneficiary of redirected US demand. If Vietnamese and Chinese products are substitutes, Vietnam may capture US market share lost by China due to tariffs. However, if Vietnam supplies intermediate goods for China's US-bound exports within global value chains, tariffs could harm its economy. Alternatively, MNCs relocating production to Vietnam to circumvent tariffs could boost its exports, as evidenced by firms like Samsung and Intel expanding operations in Vietnam. These competing dynamics necessitate empirical analysis of their impact on third countries like Vietnam.

To guide our analysis of Vietnam's exports to the United States using quarterly HS 8-digit data from 2017Q1 to 2020Q4 (with major tariff impositions in 2018Q3 and 2019Q3), we pose the following research questions. First, we ask whether Vietnam benefited from the US–China trade war through increased exports to the US. If Vietnamese goods substituted for Chinese products in the US market, we would expect a post-tariff rise in Vietnam's exports. This analysis allows us to assess the average extent of trade diversion from China to Vietnam. Second, we investigate sectoral heterogeneity by incorporating pre-war revealed comparative advantage (RCA). Specifically, we

examine whether Vietnam's export growth to the US was concentrated in sectors where (i) China held a high pre-war RCA (consistent with trade diversion and substitution effects, whereby Vietnam captured market share previously supplied by China), or (ii) Vietnam itself held a high pre-war RCA. The latter case suggests that trade-war-induced gains were attenuated in sectors where Vietnam was already competitive—likely due to capacity constraints, market saturation, or limited remaining growth potential in established export strongholds. Consequently, the largest relative gains occurred in sectors where Vietnam initially had low comparative advantage, reflecting previously untapped export opportunities. We further explore nonlinearity in substitution effects by interacting China's RCA with Vietnam's RCA, which reveals whether the substitution benefit is amplified or diminished in sectors where Vietnam was already competitive pre-war. Finally, we examine the role of MNCs in mediating these trade shifts. Discriminatory tariffs may have prompted MNCs to relocate production to Vietnam, thereby boosting its US-bound exports.[1] We expect sectors with stronger MNC presence to exhibit higher export growth to the US following the trade war.

Our findings, derived from the event study, two-way fixed effects (TWFE), and imputation-based difference-in-differences (DID) methods (Borusyak et al., 2024), reveal several insights. First, US imports from China indeed decreased significantly by approximately 32%, while imports from Vietnam increased by about 7%, though the latter is not statistically significant. The corresponding event study plot reveals a notable rise in Vietnam's exports to the US starting from the first quarter of 2020. The negative estimate on US imports from China and the positive estimate from Vietnam are consistent with the findings from the trade war literature. Second, the RCA analysis indicates that the sectors in which China had the RCA over Vietnam before the trade war experienced higher growth in Vietnam's exports to the US, suggesting that Vietnam captured part of China's exports in response to the tariffs. Third, in sectors where Vietnam already possessed a pre-war comparative advantage, the trade-war-induced export gains were smaller (or relatively attenuated) while the largest gains occurred in sectors where Vietnam initially had low comparative advantage. Fourth, and most importantly, the substitution effect exhibits nonlinearity: the positive

[1] For instance, Taiwanese firms shifted tablet and smartphone assembly for Apple's supply chain to Vietnam, while existing investors such as Samsung and Intel expanded operations there. A Nikkei survey indicates that more than 50 global companies announced or considered relocating production from China to Southeast Asia, primarily to circumvent tariffs Nikkei (2019, July 18th).

trade-diversion benefit associated with China's tariffs was weaker in sectors where Vietnam itself held a strong pre-war RCA. Overall, Vietnam captured economically meaningful trade-diversion gains from the US–China trade war, but these gains were concentrated primarily in sectors outside its pre-existing areas of comparative strength. Fifth, the surge in exports is strongly associated with heightened MNC activity. This highlights the role of MNCs in relocating their operations to Vietnam to navigate the shifting trade barriers between the US and China.

After the initial round of tariffs in the third quarter of 2018, there were continuous threats that additional tariffs could be imposed on more products. In line with this, further tariffs were, in fact, imposed on additional products in the third quarter of 2019. This raises concerns about whether the products affected by the 2019 Q3 tariffs should also be considered part of the treatment group from that point onward. Perhaps more importantly, the outbreak of the COVID-19 pandemic in early 2020 brought a significant external shock to global trade. To address these issues, we conducted a robustness analysis excluding products tariffed in 2019 Q3 and all 2020 data, focusing solely on 2018 Q3-tariffed products from 2017 Q1 to 2019 Q4. The results remain consistent with the main analysis.

Beyond the case study of Vietnam's US exports, our analysis extends to the broader impact of US tariffs on total US imports, addressing whether the significant decline in imports from China was offset by increased imports from other trading partners, including Vietnam. While our heterogeneity analyses demonstrate that Vietnam partially filled this gap by increasing its US exports, particularly in sectors with pre-war Chinese RCA and active MNC involvement, we ask to what extent imports from the rest of the world (RoW) offset the reduction in Chinese imports from the US perspective. To address this, we estimate the effect of US tariffs on total US imports, covering all trading partners. This analysis shifts the focus from country-specific effects to the aggregate impact on US imports. The results show statistically insignificant coefficients, suggesting that total US imports remained largely unchanged post-trade war. These findings have important implications for understanding the global trade reallocation triggered by the US-China trade war. The insignificant effect on total US imports suggests that US importers adapted to the tariffs by diversifying their import sources.

The paper is organized as follows: Section 2 reviews the relevant literature, Section 3 pro-

vides an overview of the trade war, Section 4 describes the data and summary statistics, Section 5 presents the methodology, empirical results, and robustness checks, and Section 6 offers concluding remarks and policy implications.

2 Literature review

The US-China trade war has been extensively studied for its impacts on prices, welfare, and trade flows in the directly affected economies. Amiti et al. (2019) provide early evidence that these tariffs led to substantial increases in US prices for intermediates and final goods, with complete pass-through to domestic consumers and importers. Their analysis highlights disruptions to supply chains, reduced import varieties, and retaliatory tariffs from China that similarly lowered real income in both countries. Extending this, Fajgelbaum et al. (2024) quantify the broader effects, finding that the tariffs caused significant declines in US imports from China and exports to China, yielding net losses to US consumers and importers, though domestic producers gained modestly. They emphasize that the trade war not only reduced bilateral trade but also prompted global reallocations, with "bystander" countries like Vietnam gaining export opportunities in tariff-exposed products.

Research on the spillover effects to third countries reveals a complex pattern of trade reallocation and supply chain adjustments. Fajgelbaum et al. (2024) show that countries with exports substituting for those from China or the US, particularly those operating along downward-sloping supply curves, benefited more from the trade war, while those with complementary exports saw smaller gains. Freund et al. (2024) examine US import data from 2017 to 2022, documenting a decoupling in US-China bilateral trade. They find that imports shifted to large developing countries with comparative advantages in affected products, often those deeply integrated into China's supply chains, especially in strategic industries. This suggests that displacing China as an exporter requires embracing its supply chains, with evidence of nearshoring to Mexico but limited reshoring to the US. Alfaro and Chor (2023) provide a long-run perspective on US global value chain participation, noting a "great reallocation" post-2017: direct US sourcing from China decreased, favoring low-wage locations like Vietnam and nearshoring options like Mexico. They also caution that such shifts may not reduce US dependence on China-linked chains and could raise import prices from

alternatives.

Studies focusing on Vietnam, a key bystander economy, highlight its gains from trade diversion amid the US-China tensions. Kahn et al. (2024) link the trade war to accelerated urban economic growth in northern Vietnam, where cities closer to China in distance and industrial mix experienced faster growth and FDI inflows. This is complemented by Nguyen and Lim (2023), who argue that the trade war unexpectedly catalyzed a broader structural transformation in Vietnam. They find that the reallocation of FDI into sectors exposed to US-China tariffs not only boosted capital inflows but also shifted the economic structure toward higher-value manufacturing, demonstrating that trade protectionism can foster industrial development in third-party developing nations.

The labor market impacts of these shifts reveal a move toward formalization and upgrading. Pham and Wie (2023) show that tariff advantages reduced informal and uninsured employment, as firms viewed the shock as a permanent demand boost. Rotunno et al. (2024) and Mayr-Dorn et al. (2023) document job creation and wage increases, particularly for women and non-college-educated workers, in districts exposed to the trade diversion. Malesky and Mosley (2021) identify a market-driven labor upgrading mechanism: foreign firms in Vietnam, particularly those in tariff-targeted industries, significantly increased their labor expenditures and improved working conditions. They argue that as these firms sought to capture new US market opportunities, the need to recruit and retain skilled labor to produce more sophisticated products incentivized voluntary improvements in wages and benefits.

Vietnam's deepened role in regional value chains has also received attention. Wu (2023) finds that foreign-owned firms, especially Chinese-invested ones, drove the surge in input sourcing and exports during the 2017–2019 period. This is further examined by Hayakawa (2024), who uses regional trade data to map the input-output linkages between US-bound exports and China-sourced imports. He finds that the rise in Vietnam's downstream exports to the US significantly increased the import of corresponding upstream intermediate goods from China, particularly in the machinery and electrical sectors. This linkage was most pronounced in provinces with superior business environments, suggesting that Vietnam's gains are highly dependent on its ability to function as a bridge for Chinese-linked supply chains.

While these shifts have benefited the Vietnamese economy, they have also introduced risks related to trade integrity and protectionist backlash. Iyoha et al. (2024) quantify trade rerouting, estimating that 1.8% to 16.1% of Vietnamese US exports in 2021 may be rerouted to Chinese goods. Xue (2024) and Graziano et al. (2024) emphasize that while FDI diversion to Vietnam helps bypass barriers, it creates a reallocation that may not reduce overall US dependence on China-linked chains. Placing these dynamics in a longer-run context, McCaig et al. (2022) note that Vietnam's capacity for export growth extends back to the US-Vietnam Bilateral Trade Agreement, but the current trade war has uniquely accelerated the intensive and extensive margins of this participation.

While existing studies document Vietnam's overall gains from the 2018–2019 US–China trade war, they largely overlook the role of pre-war RCAs of China and Vietnam. This paper fills this gap using HS 8-digit US import data (2017Q1–2020Q4) to quantify these channels. Our analysis reveals heterogeneous spillover effects and nonlinear substitution patterns, showing that Vietnam's trade-diversion gains were concentrated in sectors outside its pre-existing comparative strengths. By highlighting the interplay of comparative advantage, supply-chain linkages, and MNC relocation, we provide new insights into how bystander economies navigate major-power trade conflicts and global value chain reconfiguration.

3 Overview of trade war

President Trump's criticisms of US trade policy, particularly toward China, were a consistent theme throughout his campaign and presidency. From 2018, his administration implemented a series of protectionist trade policies, initiating the US-China trade war with tariffs on Chinese imports.

On July 6, the US implemented the first China-specific tariffs and began collecting 25 percent tariffs on $34 billion worth of imports. The US implemented a second round of tariffs on August 23, and 25 percent tariffs on another $16 billion worth of imports were imposed. The third round of tariffs was implemented on September 24. The tariffs carry an initial rate of 10 percent on an additional $200 billion worth of imports, and it was also announced that the rate would be

increased to 25 percent from January 1, 2019. The three rounds of tariffs in 2018 bring the total amount to $250 billion' worth of imports, which is almost half of the total US imports from China in the manufacturing sector.

In response, China's retaliatory tariffs covered 50 billion dollars worth of imports from the US: China's new 25 percent tariffs were implemented in July and August, covering 34 billion dollars and 16 billion dollars of imports. China unleashed another round of tariffs in September, imposing new 5 percent and 10 percent tariffs on 60 billion dollars' worth of imports from the United States.

The war continued in 2019. On May 10, the US increased the third round of tariffs from 10 percent to 25 percent, which was initially planned for January 1, 2019. On September 1, the US began implementing the fourth round of tariffs: 15 percent tariffs on an additional 125 billion dollars worth of imports. It was also announced that a new round of tariffs on 175 billion dollars worth of imports would be implemented from December 15, which would have resulted in all imports from China being subject to tariffs.

In response to these tariff hikes, in June 2019, China also raised its tariff rates on the product list that had already been targeted in September 2018, covering $36 billion worth of imports. China also announced tariffs covering 75 billion dollars worth of imports to be imposed in September and December 2019, with the December list being later canceled.

The conflicts ended temporarily in January 2020 when both sides reached the Phase One deal. Through this deal, tariffs planned to go into effect in December were canceled, and tariffs that began in September 2019 were reduced from 15 percent to 7.5 percent. China also pledged to boost imports from the US and strengthen IP rules. In total, at this point, the US tariffs on Chinese products cover a list representing 362 billion dollars worth (in terms of their 2017 value), which represents roughly 72% of the imports from China in 2017. Chinese tariffs apply to a list of products representing about 97% of US exports to China in 2017.

4 Data and descriptive statistics

We obtain quarterly US import data from China and Vietnam via the USITC, using the HS8-digit classification. Our sample covers the period from the first quarter of 2017 to the last quarter

of 2020, with tariff impositions in 2018Q3 and 2019Q3. Our analysis focuses on the manufacturing sector and retains only products exported by China and Vietnam to the US at least once during the sample period between 2017Q1 and 2020Q4. This selection yields 4,597 unique products. We then construct a balanced panel of 16 quarters, resulting in 73,552 observations. In cases where the export values are zero, we first add one and then apply the natural logarithm transformation.[2]

The US tariff schedule from the USITC specifies when tariffs were imposed based on HS8-digit classification, categorized under five codes: 9903.88.01, 9903.88.02, 9903.88.03, 9903.88.04 (introduced in 2018Q3), and 9903.88.15 (introduced in 2019Q3). Of the 4,597 products, 14.9% were exempt from tariffs, serving as the control group. 56.9% were tariffed in 2018Q3 and the remaining 29% received tariffs in 2019Q3, reflecting the staggered nature of the trade war's policy shocks. These tariff assignments enable us to isolate the treatment effect of tariffs on US imports from China and Vietnam, accounting for product-level heterogeneity in tariff exposure.

In order to explore sectoral heterogeneity, we examine the RCA of China and Vietnam in their exports to the US. Following the standard definition from Balassa (1965), we calculate the RCA index using 2017 observations as follows:

$$RCA_i = \frac{\frac{\text{China's (or Vietnam's) exports to the US of a good } i}{\text{China's (or Vietnam's) total exports to the US}}}{\frac{\text{RoW's exports to the US of a good } i}{\text{RoW's total exports to the US}}}.$$

This measure captures China's and Vietnam's pre-war competitive position in the US market, where higher RCA values indicate a stronger comparative advantage in 2017.

Using China's RCA, we investigate whether Vietnam's export growth to the US following the trade war was concentrated in sectors where China previously held a comparative advantage—consistent with substitution effects. Using Vietnam's own RCA, we assess whether Vietnam's post-war export growth occurred primarily in sectors where Vietnam already held a comparative advantage or in sectors where it did not. The former would suggest that Vietnam expanded exports of products in which it was already competitive, while the latter would indicate that Vietnam successfully entered or expanded into products outside its pre-existing areas of

[2]We also employed the inverse hyperbolic sine instead of the log value. The inverse hyperbolic sine of some variable, x, is given by $ln[x + (x^2 + 1)^{0.5}]$. It equals 0 when $x = 0$, and its slope tracks that of $ln(x)$ more closely than $ln(1 + x)$ when x is small. The results remain consistent with those using the logarithmic transformation.

comparative advantage. More importantly, we examine potential non-linear effects by interacting China's RCA with Vietnam's RCA. If substitution effects are present—i.e., Vietnam exports more in sectors where China held a comparative advantage—the interaction term between China's RCA and Vietnam's RCA will reveal whether this substitution effect becomes weaker or stronger in sectors where Vietnam itself also possessed a comparative advantage.

We also specifically assess the role of MNCs in driving Vietnam's export growth, as discriminatory tariffs may have incentivized production relocation to Vietnam. MNC activity is measured as the share of US related-party imports from Vietnam for product i:

$$MNC_i = \frac{\text{US related party import of a good } i \text{ from Vietnam}}{\text{US total import of a good } i \text{ from Vietnam}}$$

The data is extracted from the US Census Bureau based on NAICS 6-digit classification.[3]. To assess MNC activity in Vietnam, we construct a dummy that identifies when product-i is imported to the US from a MNC.

Table 1 is the summary statistics of our data. The impact of tariffs is evident, as the log value of China's exports to the US decreased, while Vietnam's exports increased after the tariff imposition. Figures 1 and 2 illustrate these shifts. Figure 1 depicts the export values to the US, while Figure 2 shows the number of HS8-digit products exported. Both figures highlight a sharp decline in China's exports alongside a steady rise in Vietnam's exports throughout the trade war.

The average RCA values for China and Vietnam are 1.242 and 1.081, respectively, indicating that both countries exhibited a strong revealed comparative advantage relative to the RoW in the US market prior to the trade war. The mean value of the MNC activity indicator is 0.725, suggesting that multinational corporation activity was present in 72.5% of the Vietnamese sectors included in our sample following the trade war. Notably, the large standard deviations, particularly for the RCA measures and the MNC activity indicator, point to substantial heterogeneity across products (or sectors). The following subsection examines this sectoral heterogeneity in greater detail.

Table 2 provides the number of the unique products exported from China and Vietnam to the US during the sample period, offering a detailed view complementing Figure 2. The first

[3]To match the NAICS and HS codes, we follow the NAICS-HS concordance obtained by (Liao et al. (2020))

row displays the number of products exported by both countries to the US, showing a steady increase over time. The second row shows that the number of products exported solely by China has decreased, while the third row indicates that the number of products exported exclusively by Vietnam has remained constant. Together, these rows suggest that Vietnam has begun to export additional products that China previously exported. Additionally, the fourth row lists the number of products not exported by either country in that year but which were exported at least once during the sample period.

5 Methods and results

In this section, we conduct a regression analysis of the trade war's impact. We first present the average treatment effect estimates based on two-way fixed effects (TWFE) and a recently developed imputation-based method, followed by robustness checks. Next, we examine sectoral heterogeneity in China's RCA, Vietnam's RCA, and MNC activity. We replicate these analyses with a restricted sample (2017Q1–2019Q4, with 2018Q3 tariffs) to address the concerns regarding 2019Q3 tariffs and COVID-19 disruptions. Finally, we assess the trade war's overall effect on total US imports.

5.1 Average Treatment Effect

Before presenting the average treatment effect estimates, we summarize the estimation procedures for the TWFE and imputation-based difference-in-differences (DID). The TWFE DID regression is specified as:

$$Y_{it} = \tau D_{it} + \alpha_i + \delta_t + \epsilon_{it} \tag{1}$$

where Y_{it} is the outcome, D_{it} is the treatment indicator, and α_i and δ_t are individual and time fixed effects, respectively. In this study, the outcome is the exports from Vietnam to the US and the treatment occurs either 2018Q3 or 2019Q3.

To address staggered treatment timing in our analysis of US tariff impacts (2018Q3 and 2019Q3), we adopt the imputation-based DID method proposed by Borusyak et al. (2024), which offers robustness over traditional TWFE models. This approach proceeds as follows: First, we estimate

product and time fixed effects, $\hat{\alpha}_i$ and $\hat{\delta}_t$, using the regression:

$$Y_{it} = \alpha_i + \delta_t + \beta X_{it} + \epsilon_{it} \tag{2}$$

where the sample includes only untreated units (products exempt from tariffs) or not-yet-treated units (pre-tariff periods). Using these estimates, we compute counterfactual outcomes, $\hat{Y_{it}(0)}$, for treated units. The treatment effect for each treated observation is calculated as the difference between the observed outcome, Y_{it}, and the counterfactual, $\hat{Y_{it}(0)}$. These effects are aggregated via weighted averages to estimate the average treatment effect. This method is efficient under limited serial correlation in outcomes and plausible parallel trends, which is supported by pre-treatment trends in our data. It is particularly suited to our study due to the staggered tariff implementation, ensuring robust estimation of trade war effects. For details, see Borusyak et al. (2024).

The estimation results reported in Table 3 confirm a negative treatment effect on China's exports to the US under both TWFE and imputation-based methods. The imputation-based estimate indicates that US tariffs reduced China's exports by approximately 32%. The corresponding event-study plot (Figure 3) shows that the tariffs had an immediate impact, with a significant and negative coefficient at the treatment date (2018Q3 and 2019Q3). Over time, the effect generally becomes increasingly negative, suggesting a sustained impact. Notably, two positive data points before the tariffs suggest that China may have accelerated exports to the US in anticipation of the policy, especially given that the tariff imposition in the 3rd quarter of 2019 was much anticipated.

For Vietnam's exports to the US, the results indicate a modest increase, though neither method yields statistically significant effects, with the imputation-based estimate indicating a 7% rise. The event-study plot, however, reveals a gradual increase starting six quarters post-2018Q3, suggesting a lagged response. This gradual response likely reflects Vietnam's increasing comparative advantage and the influence of MNCs, which we will explore in the next subsection, suggesting that the tariff-induced shift in trade patterns unfolds gradually over time rather than immediately.

5.2 Robustness Check

To ensure the robustness of our imputation-based DID estimates, we conduct a permutation test, which validates the statistical significance of the average treatment effects reported in Table 3. The permutation test assesses whether our results for China and Vietnam's US exports are driven by the actual tariff treatment schedule (2018Q3 and 2019Q3) or could arise by chance under random assignment.

The permutation test proceeds as follows. We randomly reassign tariff imposition dates (2018Q3 or 2019Q3) across the 4,597 HS8-digit products in our sample, preserving the original treatment structure, including the proportion of products treated in each period (56.3% in 2018Q3, 28.6% in 2019Q3, and approximately 15% untreated). This shuffling maintains the temporal distribution of tariffs and the number of treated observations, ensuring comparability with the actual treatment schedule. We perform 1,000 such permutations, re-estimating the ATE for each using the imputation-based DID method outlined in Section 4.1. For each permutation, we compute the t-statistic of the ATE, creating a distribution of permuted t-statistics. We then compare this distribution to the original t-statistics from Table 3 to assess the likelihood of observing our results under random assignment. A result is deemed statistically significant at the 5% level if fewer than 50 of the 1,000 permuted t-statistics exceed the original t-statistic in absolute value, indicating that the observed ATE is unlikely to be a random outcome.

The results of the permutation test support the robustness of our findings. For China, the original t-statistic from the imputation-based DID method is approximately -5.745, reflecting a significant 32% reduction in US exports (Table 3). Figure 5 plots the distribution of permuted t-statistics for China's ATE after 1,000 iterations, showing that no permuted t-statistic exceeds -5.745 in absolute value. This confirms that the negative ATE for China is highly statistically significant, as the observed effect is not replicable under random treatment assignments. For Vietnam, the original t-statistic is 0.683, corresponding to an insignificant 7% increase in US exports. Figure 6 displays the permuted t-statistic distribution for Vietnam's ATE, with a vertical line marking the original value of 0.683. The distribution shows that many permuted t-statistics exceed this value, consistent with the insignificance of Vietnam's ATE. These permutation tests reinforce the reliability of our imputation-based estimates, ensuring that the significant effect for China and the

insignificant effect for Vietnam represent the true impact of the US tariffs rather than spurious correlations.

5.3 Heterogeneity

To explore sectoral heterogeneity, we rely on the TWFE model by incorporating interaction terms that capture sectoral characteristics. This approach allows us to test whether Vietnam's export growth to the US is concentrated in sectors where China held an RCA. or where Vietnam itself held an RCA. We further examine non-linear effects by interacting China's RCA with Vietnam's RCA, which reveals whether substitution effects are stronger or weaker in sectors where Vietnam is already competitive. Finally, we assess whether Vietnam's export expansion is concentrated in the MNC active sectors. The model is specified as:

$$Y_{it} = \tau_1 D_{it} + \boldsymbol{\tau}_2 D_{it} * X_i + \alpha_i + \delta_t + \epsilon_{it} \tag{3}$$

The vector X_i includes the following sectoral characteristics: (1) China's RCA in exports to the US relative to the rest of the world, measured using 2017 data; (2) Vietnam's RCA in exports to the US relative to the rest of the world, measured using 2017 data; (3) the interaction between China's RCA and Vietnam's RCA; and (4) an indicator for positive MNC activity in Vietnam post-trade war. The vector $\boldsymbol{\tau}_2 = (\tau_{2,\text{China}}, \tau_{2,\text{Vietnam}}, \tau_{2,\text{interaction}}, \tau_{2,\text{MNC}})'$ captures the heterogeneous treatment effects, showing how the impact of US tariffs on Vietnam's exports varies with each sectoral characteristic.

Table 4 presents the main results. Column (1) reports the baseline TWFE estimates for Vietnam, previously shown in Table 3. Column (2) includes interactions with China's pre-war RCA, Vietnam's pre-war RCA, and their triple interaction. The significantly positive coefficient on China's RCA indicates that Vietnam experienced stronger export growth to the US in sectors where China held an RCA prior to the trade war. This pattern is consistent with trade diversion and substitution effects, whereby Vietnam captured market share previously held by China. In contrast, the significantly negative coefficient on Vietnam's own RCA suggests that the trade-war-induced export gains were smaller (or relatively attenuated) in sectors where Vietnam already possessed a pre-war comparative advantage—likely reflecting capacity constraints, market saturation, or lim-

ited remaining growth potential in established export strongholds. The largest gains, therefore, occurred in sectors where Vietnam initially had low comparative advantage, reflecting previously untapped export opportunities. Most importantly, the negative and significant coefficient on the triple interaction term (China RCA $\times$ Vietnam RCA $\times$ treatment) reveals a non-linear pattern in the substitution effect. The positive trade-diversion benefit from China's tariffs (β_2) was substantially weaker in sectors where Vietnam already held a strong pre-war RCA. The results suggest that Vietnam captured meaningful trade-diversion gains from the US-China trade war, but primarily in sectors outside its pre-existing areas of comparative advantage, where export opportunities were previously untapped. In industries characterized by high overlap (and thus high values of both RCAs), Vietnam often served as a downstream assembler, relying on Chinese intermediate inputs. Disruptions to these regional production networks during the trade war likely limited Vietnam's ability to expand exports, precisely in those industries.

The heterogeneous treatment effects reveal economically meaningful magnitudes. Evaluated at the sample means of China's pre-war RCA (1.242) and Vietnam's pre-war RCA (1.081), the net marginal effect of the US–China trade war on Vietnam's exports to the US is approximately 0.24%. This modest change masks substantial heterogeneity, however. Compared to sectors where China had no revealed comparative advantage (China RCA = 0), Vietnam's exports increased by an additional 16.6% when China's RCA rises from 0 to its mean value (holding Vietnam RCA at its mean). This quantifies the strength of the trade-diversion (substitution) channel. Conversely, moving Vietnam's own RCA from its mean (1.081) down to 0—while holding China RCA at its mean—boosts the trade-war effect by an additional 20.3%, illustrating that the largest relative gains occurred in previously untapped or weakly competitive sectors.

In Column (3), we examine the role of MNCs. Anecdotal evidence suggests that many global companies, previously operating subsidiaries in China, have shifted their supply chains to other countries to avoid US tariffs.[4] Vietnam presents an attractive alternative for these companies, not only due to its geographical proximity to China but also because of its lower labor costs. Column (3) tests whether MNC activity in Vietnam after the trade war has driven its post-trade war exports to the US, yielding statistically significant results. These findings underscore the critical role of

[4]For further reading, see related articles by (BBC, 2019, May 19th; Nikkei, 2019, July 18th).

global value chain reallocation in amplifying Vietnam's trade response to the US-China trade war.

Column (4) combines all interaction terms, confirming the roles of RCAs and MNC for Vietnam's exports, with consistent signs and significance. Coefficient magnitudes vary slightly due to correlations among X_i variables, but the overall interpretation holds.

In summary, the analysis of sectoral heterogeneity shows that Vietnam's post-war export growth to the US was shaped by pre-war RCAs of both China and Vietnam, as well as MNC relocations. The results underscore the interplay of trade substitution, tariff circumvention, and global value chain restructuring, illustrating how third countries can serve as critical intermediaries that absorb and redirect trade flows during bilateral disruptions between major economies.

5.4 Restricted Model

To ensure the robustness of our findings against potential confounding factors, we conduct a restricted sample analysis that excludes products tariffed in 2019Q3 and data from 2020, addressing two key concerns. First, following the initial US tariffs on Chinese imports in 2018Q3, ongoing policy announcements signaled potential additional tariffs, which materialized in 2019Q3, affecting 1,286 products (29% of the sample). These later tariffs may have induced anticipatory trade adjustments, complicating the identification of 2019Q3 tariff effects. Second, the COVID-19 pandemic in 2020 disrupted global trade, introducing demand and supply shocks that could obscure tariff impacts. If these shocks are uniform across sectors, time fixed effects can account for them. However, sector-specific shocks, which are more likely, complicate isolating tariff impacts. To address this, we exclude 2019Q3-tariffed products and the observations from the year 2020. By doing so, the focus is purely set on the 2018Q3 tariffs (2,525 products, 56.9% of the sample). Given the single treatment timing (2018Q3), we employ a traditional TWFE model.

The TWFE model is specified as in Equation 3, with D_{it} as the treatment indicator for 2018Q3 tariffs. The restricted sample includes 3,283 products over 12 quarters (2017Q1–2019Q4). This approach ensures a clean identification of the 2018Q3 tariff effects, leveraging the 691 non-tariffed products as the control group.

Table 5 reports the regression results for the restricted sample. These findings are broadly consistent with the full-sample results, albeit with reduced statistical significance. In Column (1), the

average treatment effect for Vietnam's exports to the US is positive but statistically insignificant, mirroring the full-sample ATE. Column (2) reveals a positive interaction with China's 2017 RCA, suggesting export growth in sectors where China previously held a comparative advantage while this estimate is not statistically significant. Consistent with Table 4, the negative and significant coefficient on Vietnam's 2017 RCA indicates that sectors with a strong pre-war export presence did not drive the subsequent growth. Instead, export expansion was primarily concentrated in sectors where Vietnam initially held little comparative advantage. Furthermore, the negative and significant coefficient on the triple interaction indicates the non-linearity in the substitution effects. Finally, the positive and significant MNC coefficient in Column (3) points to a relocation of global value chains toward Vietnam. Column (4) incorporates all interaction terms simultaneously, yielding consistent results.

In summary, the restricted sample analysis reinforces the core findings: Vietnam's export surge following the 2018Q3 tariffs was driven by shifts in China's RCA, growth in sectors where Vietnam previously lacked a comparative advantage, and MNC-led relocation. The consistency of these results across the restricted time frame mitigates concerns regarding anticipatory behavior or COVID-19-induced disruptions, underscoring the robustness of the identified mechanisms.

5.5 Overall Effect of the Trade War on US Imports

This subsection extends our analysis beyond Vietnam's US exports to evaluate the broader impact of the US-China trade war on total US imports, addressing the following question: did increased imports from other trading partners, including Vietnam, offset the significant decline in imports from China? The trade war, initiated with US tariffs on Chinese goods in Q3 2018 and Q3 2019, reduced China's exports to the US by approximately 32%, disrupting bilateral trade flows. Our heterogeneity results demonstrate that Vietnam partially compensated for this decline, particularly in sectors with high pre-war Chinese RCA and MNC activity. However, the extent to which the RoW collectively mitigated the reduction in Chinese imports remains unclear, with implications for understanding global trade relocation and US import resilience.

To address this, we estimate the effect of US tariffs on total US imports using HS8-digit quarterly data from 2017Q1 to 2020Q4, covering all trading partners. Unlike the country-specific anal-

yses, which focus on China and Vietnam, this approach captures the net impact on the US imports by summing imports for each product across all origins.

The results, reported in Table 6, provide consistent evidence across all specifications. Columns (1) and (2) respectively present TWFE estimates for the entire and the restricted sample, while Column (3) shows imputation-based DID results. All three columns yield positive but statistically insignificant coefficients, suggesting that total US imports remained largely stable post-trade war, with no significant net decline despite China's substantial export reduction. US importers diversified their sources to compensate for reduced Chinese imports.

The findings highlight the resilience of US import markets and the pivotal role of third countries in global value chain relocation. Vietnam's proximity to China and low labor costs positioned it as a key beneficiary, but the collective contribution of RoW countries ensured minimal disruption to US import volumes. These results underscore the limitations of discriminatory protectionist tariffs, as global trade networks adapt through diversification.

6 Concluding Remarks

We examine how the US-China trade war influenced Vietnam's US exports through the lenses of sectoral heterogeneity and GVC relocation. Analyzing HS8-digit US import data (2017Q1–2020Q4), we find that a 7% increase in Vietnam's US exports post-trade war is masked by an insignificant average effect. A further analysis, however, identifies robust growth driven by China's prior comparative advantage, Vietnam's expansion into low-RCA sectors, and heightened MNC activity. Additionally, our analysis of total US imports reveals no significant change, suggesting that the 33% decline in China's US exports was fully offset by increased imports from the RoW, including Vietnam.

While our results position Vietnam as a beneficiary of the trade war, the generalization of these findings to other countries remains an open question. The trade war's unintended consequences likely depend on country-specific factors, such as the substitutability of a country's product mix for Chinese goods, the presence of MNC activity, and participation in trade agreements facilitating US market access. For instance, countries like Mexico, with a product mix overlapping

China's, significant MNC operations, and preferential access via the USMCA, likely experienced export gains similar to Vietnam's. Conversely, countries with less substitutable exports and limited MNC presence may have seen stagnant or reduced US exports. Future research could test this hypothesis by comparing export responses across a broader set of countries to identify the economic and institutional conditions that amplify or mitigate the effects of trade protectionism between major economies.

Despite the granularity of our HS8-digit data, limitations persist, particularly the inability to track firm-level responses to the 2018–2019 tariffs. An ideal analysis would leverage firm-level data to examine how characteristics such as production type (labor-intensive versus capital-intensive), firm size, or ownership structure influenced their ability to benefit from trade relocation. For instance, labor-intensive MNCs, such as those in apparel or electronics assembly, likely leveraged Vietnam's low labor costs to relocate operations and capitalize on tariff-induced opportunities more effectively than capital-intensive firms, such as those in heavy machinery, which face higher relocation costs. Investigating these firm-level dynamics would yield deeper insights into the microeconomic mechanisms driving trade war responses, illuminating how production characteristics shape global supply chain reconfigurations.

Tables

TABLE 1: Summary Statistics

Variable	Mean	Standard Deviation
Log of China's US Exports before trade war	13.686	3.969
Log of China's US Exports after trade war	13.389	4.134
Log of Vietnam's US Exports before trade war	6.098	6.221
Log of Vietnam's US Exports after trade war	7.072	6.284
China's RCA on US Exports	1.242	1.027
Vietnam's RCA on US Exports	1.081	2.878
MNC	0.725	0.446

Note: Our RCA measures are derived from 2017 observations, consistent with the specification previously outlined. For both China and Vietnam, we first calculate the export share of product i in each country's total exports to the US in 2017. We then compute the corresponding share for the rest of the world. The final RCA value is defined as the ratio of each nation's export share to that of the rest of the world. Additionally, the MNC variable is a binary indicator that equals 1 if Vietnam's related-party exports to the US are positive from 2018 to 2020, and 0 otherwise.

TABLE 2: The Number of Export Products

Export Type / Year	2017	2018	2019	2020
Both China and Vietnam Exported to the US	3,120	3,207	3,493	3,635
Only China Exported to the US	1,401	1,313	1,010	857
Only Vietnam Exported to the US	44	37	47	49
Neither Exported to the US	32	40	47	56

Note: The table reports the number of unique products exported from China and Vietnam to the United States. The first row presents products exported by both countries, the second row reports products exported exclusively by China, the third row reports products exported exclusively by Vietnam, and the fourth row reports products exported by neither country.

TABLE 3: Average Treatment Effect

	China		Vietnam	
	TWFE	Imputation	TWFE	Imputation
Treatment effect	-0.184**	-0.316***	-0.027	0.069
	(0.073)	(0.055)	(0.118)	(0.101)
Observations	73,552	73,552	73,552	73,552
Number of Goods	4,597	4,597	4,597	4,597

Note: The average treatment effect estimates for US imports from China (left) and Vietnam (right) are reported using TWFE and Imputation estimators. The dependent variable is the quarterly HS 8-digit US import value for the 2017Q1–2020Q4 period. Treatment onset occurs in 2018Q3 or 2019Q3. Robust standard errors are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

TABLE 4: Heterogeneity on Vietnam's US Exports

	(1)	(2)	(3)	(4)
Treatment Effect	-0.0266	-0.0216	-0.569***	-0.595***
	(0.119)	(0.132)	(0.220)	(0.222)
Treatment Effect × China RCA		0.181***		0.170***
		(0.0484)		(0.0486)
Treatment Effect × Vietnam RCA		-0.134***		-0.135***
		(0.0255)		(0.0255)
Treatment Effect × China RCA × Vietnam RCA		-0.0416**		-0.0419**
		(0.0192)		(0.0192)
Treatment Effect × MNC			0.559***	0.605***
			(0.192)	(0.190)
Observations	73,552	73,552	73,552	73,552
R-squared	0.033	0.038	0.033	0.038
Number of Goods	4,597	4,597	4,597	4,597

Note: This table reports TWFE estimates for US imports from Vietnam, focusing on interactions with the treatment indicator. The dependent variable is the quarterly HS 8-digit US import value spanning 2017Q1–2020Q4. Column (1) reports the baseline average treatment effect without interactions. Columns (2) and (3) introduce interaction terms for RCA and MNC, respectively. Column (4) simultaneously includes both the Treatment×RCA and Treatment×MNC interaction terms. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1

TABLE 5: Heterogeneity (Restricted Model) on Vietnam's US Exports

	(1)	(2)	(3)	(4)
Treatment Effect	0.0413	0.0640	-0.318	-0.335
	(0.116)	(0.131)	(0.223)	(0.227)
Treatment Effect × China RCA		0.0812		0.0749
		(0.0562)		(0.0564)
Treatment Effect × Vietnam RCA		-0.108***		-0.109***
		(0.0357)		(0.0358)
Treatment Effect × China RCA × Vietnam RCA		-0.0663**		-0.0666**
		(0.0290)		(0.0291)
Treatment Effect × MNC			0.373*	0.421**
			(0.203)	(0.205)
Observations	39,396	39,396	39,396	39,396
R-squared	0.027	0.029	0.027	0.029
Number of Goods	3,283	3,283	3,283	3,283

Note: The table presents TWFE regression results on interactions with the treatment variable after excluding products tariffed in 2019Q3 and data from 2020. Column (1) reports the baseline average treatment effect without interactions. Columns (2) and (3) introduce interaction terms for RCA and MNC, respectively. Column (4) simultaneously includes both the Treatment × RCA and Treatment × MNC interaction terms. Robust standard errors are reported in parenthesis. *** p<0.01, ** p<0.05, * p<0.1

TABLE 6: Aggregate Impact on US Imports

	(1)	(2)	(3)
Treatment effect	0.063	0.023	0.021
	(0.071)	(0.065)	(0.051)
Observations	73,552	39,396	73,552
Number of Goods	4,597	3,283	4,597

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: The table reports the effect of US tariffs on total US imports covering all trading partners. Columns (1) and (2) show average treatment effects based on TWFE and exclusion of products tariffed in 2019Q3 and observations from 2020. (3) presents imputation-based methods without sample restrictions. Robust standard errors are reported in parenthesis. *** p<0.01, ** p<0.05, * p<0.1

Figures

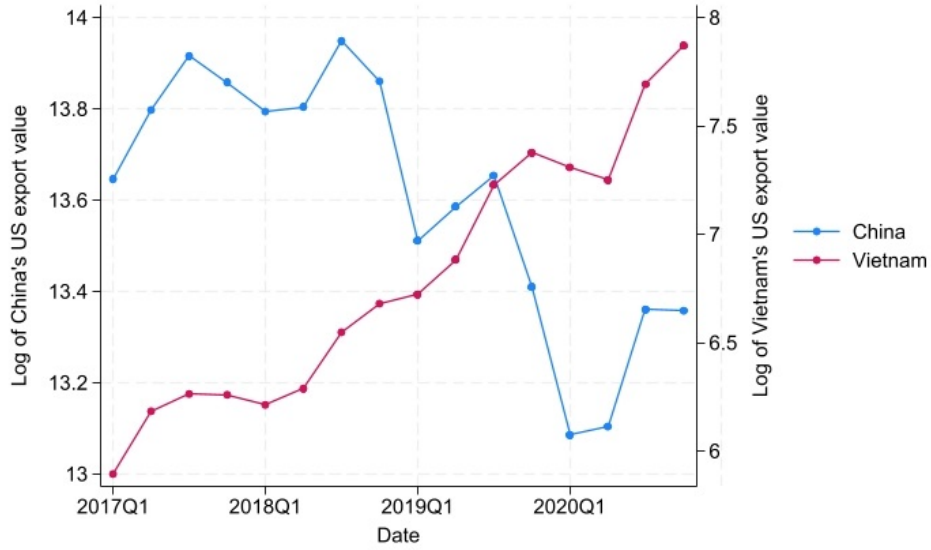

FIGURE 1: US Import Values from China and Vietnam

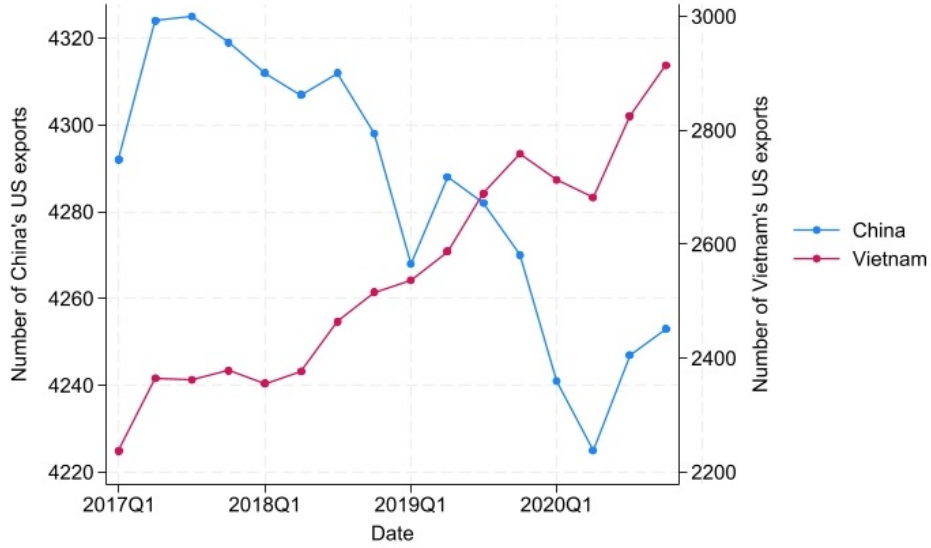

FIGURE 2: The Number of US Imports from China and Vietnam

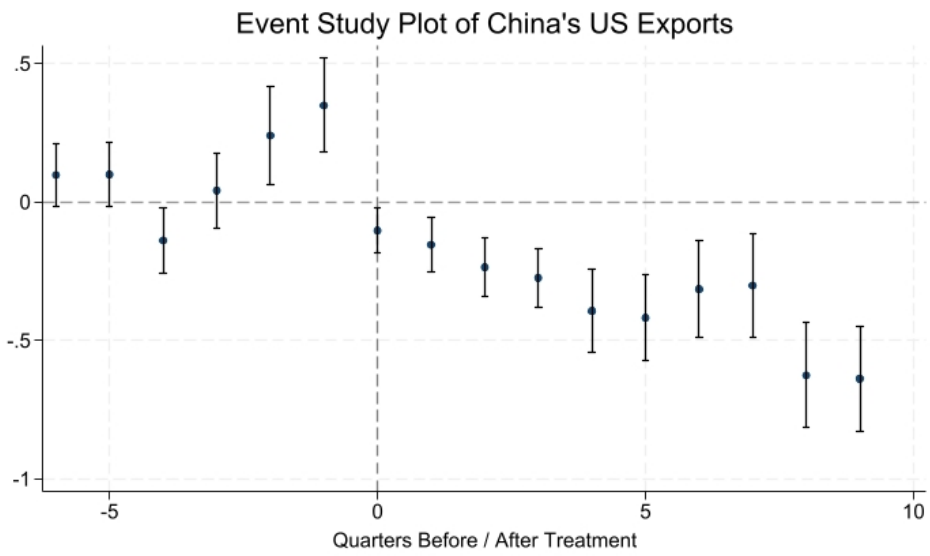

FIGURE 3: Event Study Plot of China's US Exports

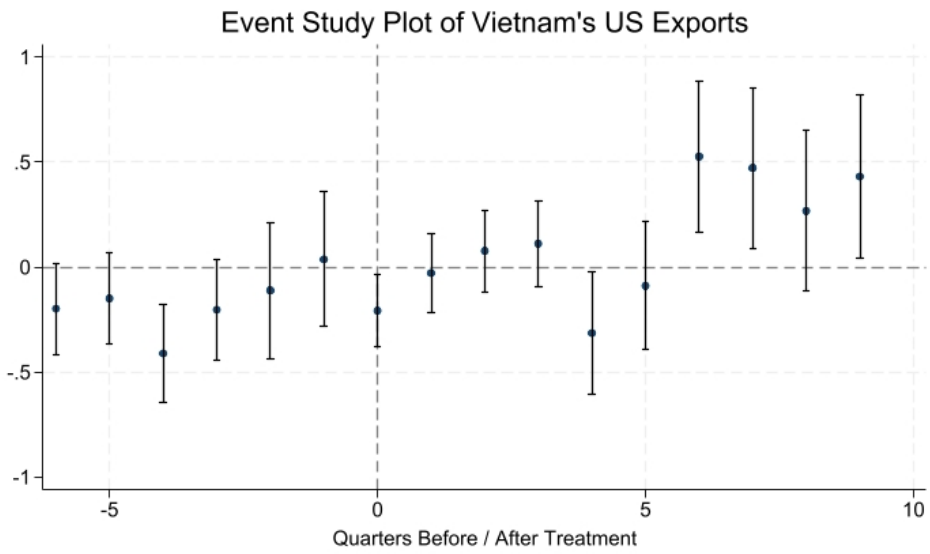

FIGURE 4: Event Study Plot of Vietnam's US Exports

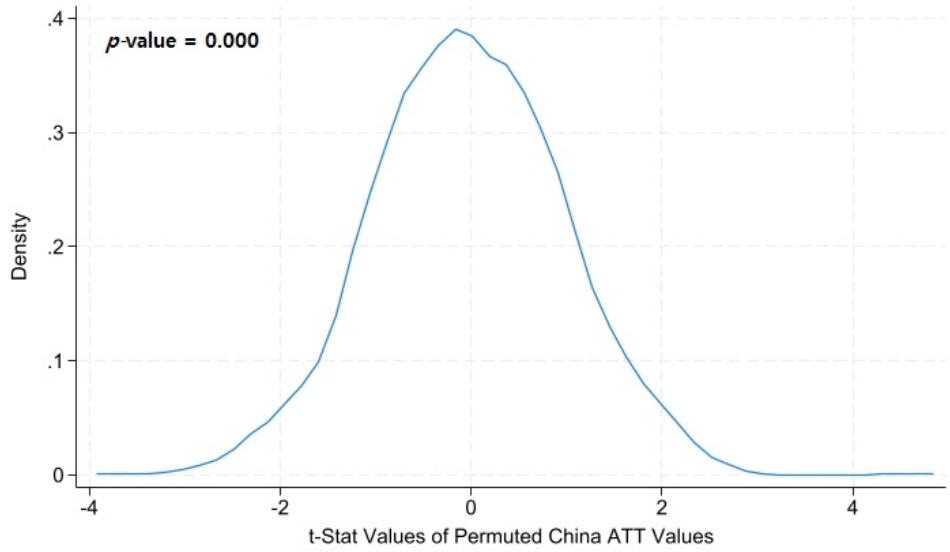

FIGURE 5: Kernel Density Graph of Permuted China's US Export ATT t-stat

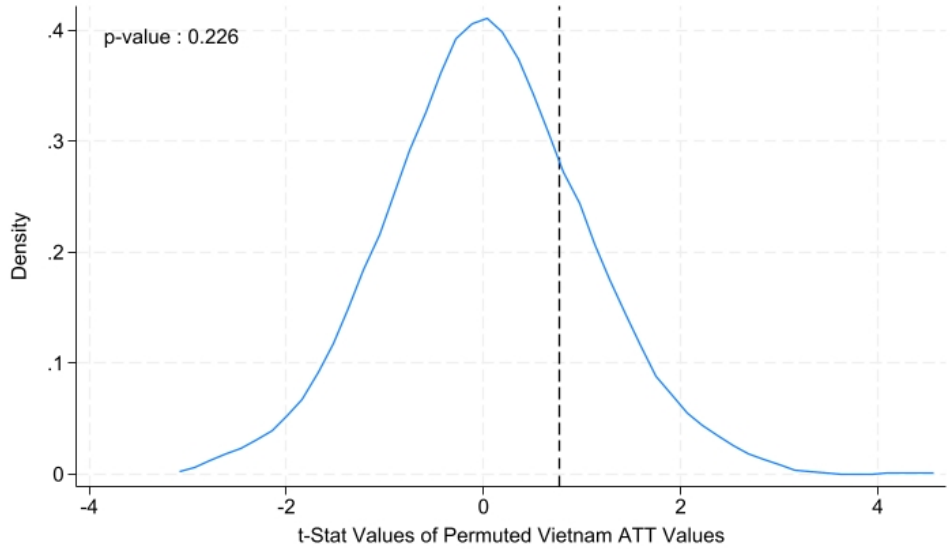

FIGURE 6: Kernel Density Graph of Permuted Vietnam's US Export ATT t-stat

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT, developed by OpenAI, for the purpose of improving the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and will take full responsibility for the content of the published article.

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